# Universality of Phase Space as Gaussian Spectrum

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# Introduction

In the study of soft condensed matter, the phase space of a many-body system plays a unique and central role. Even though the number of possible configurations grows exponentially with the system size for large-scale simulations, this apparent intractability does not render the problem hopeless; in fact, universal statistical regularities often emerge. Notably, a series of systems, including the magnetically frustrated models, one-dimensional lattice gases, and spin glasses investigated by Yilong Han and co-workers between 2009 and 2015 [6, 7, 11, 4], have revealed a remarkable universality: when one considers the connectivity graph of the phase space for their models and examines the empirical spectral distribution (ESD) of its adjacency matrix, the limiting density of states, after appropriate normalization, converges to a Gaussian distribution in the thermodynamic limit. This observation suggests that the spectral properties of such connectivity matrices are largely insensitive to microscopic details, hinting at a deeper, model-independent mechanism.

The present work aims to identify sufficient conditions under which this Gaussian universality holds. Instead of studying specific graph geometries case by case, we formulate the problem in operator algebra for a unified treatment. Our main tool is the method of moments combined with an assumption on the decay of cumulate traces. We show that if the local operators are well behaved with correlations decaying exponentially, then the ESD of the sum of these operators converges weakly to a Gaussian law. This Gaussian density is not the semicircle law familiar from sparse random graphs [12]; instead it is produced by pairing moments of sum of local operators for a connected graph, rather than by the spectral edge of a random graph.

While the argument is algebraic while the phenomenon it explains is physical, the paper is organised so that the two can be read separately. The main text states the limit theorems and defers the estimates to the appendices. Figure 1 summarises the overall storyline, from the numerical observation of Gaussian phase-space spectral densities to the operator-algebraic Gaussian limit, the conserved sectors, and the decimation fixed point.

Section 1 is both a pedagogical warm-up and the sharp limiting case against which further generalisation is measured; its elementary derivation is reproduced in Appendix B. The configuration graph of a periodic lattice gas is a hypercube in which every vertex has exactly $N$ neighbours (Figure 2). In this basic model of commute local spin-flip operators and exactly diagonalizable adjacency operator, the density of states becomes a sum of independent Rademacher variables, and the classical central limit theorem (CLT) yields a Gaussian law (Figure 3).

Section 2 generalizes features of solvable hypercube model by distant local operators whose cluster commutators decay exponentially in the diameter of their supports (Figure 4). Under such hypothesis stated in Appendix C, the ESD of the normalised global operator is again Gaussian, with a variance determined by the two-point correlations (Theorem 2.1). The locality hypothesis is stated in full in what follows, and illustrated on a toy model, in Figure 11 of Appendix C; the proof is developed in Appendix D.

Section 3 treats systems with symmetries associated with conserved charges. Upon applying statement of symmetry, the adjacency operator becomes block diagonal in the eigenspaces of the charge (Figure 5). Each sector inherits the Gaussian limit, and the full spectrum is a mixture of sector-wise Gaussians which collapses to a single Gaussian precisely when the sector variances concentrate (Figure 6). Furthermore, the same reasoning governs the degrees, or row sums, of the configuration network (Figure 7), and it survives the case in which the charge is conserved only approximately (Figure 8). The hypotheses, the discussion of when the mixture collapses,

and the proofs are collected in Appendix E.

Section 4 shows that the Gaussian limit is stable under a decimation-type coarse graining. The coarse-grained system obeys the same limiting law up to a deterministic rescaling, and the network is asymptotically self-similar (Figure 9). This is the precise sense in which the fractal structure reported by Han and co-workers is reproduced by the present framework; the estimates are given in Appendix F.

Section 5 verifies the hypotheses in Han's models [6, 7, 11, 4] that motivated this study, the frustrated spin glass of conserved energy and the lattice gas with conserved particle number (Figure 10). In both cases the interactions are strictly local, so that the locality hypothesis holds trivially and no further estimate is required.

Throughout, the hypotheses are referred to by the labels (A1)–(A5) and (C0)–(C4) attached to them in Appendices C and E, where they are stated in full, discussed, and used; all proofs are given there as well. A summary of the notation is provided in Appendix A.

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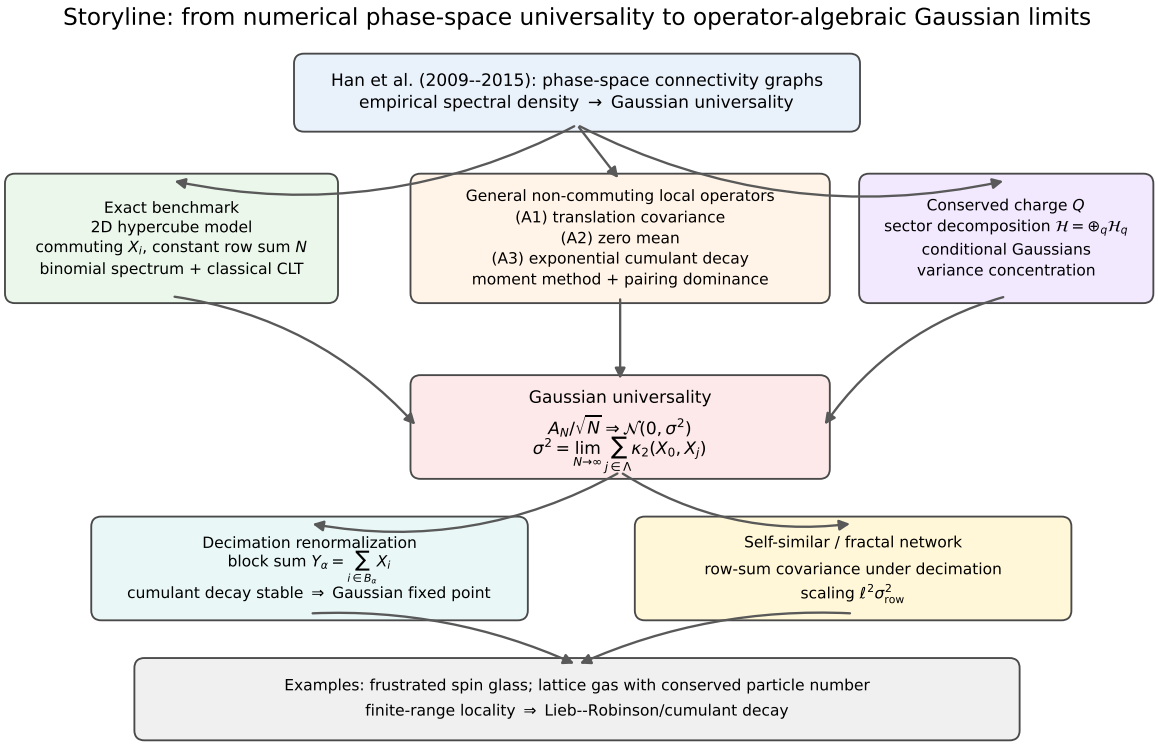



**Figure 1.** Overall storyline: from the numerical observation of Gaussian phase-space spectral densities to the operator-algebraic Gaussian limit, including conserved sectors and decimation self-similarity.

# 1 The exactly solvable hypercube limit

The natural starting point and perhaps the simplest model to verify the Gaussian law of ESD is the configuration graph of a lattice gas on a two-dimensional periodic lattice (torus) of $N = L_1 L_2$ sites. Its vertices are the $2^N$ occupation-number configurations, and two vertices are joined by an edge when they differ by a single spin flip, so that the graph is a hypercube. Every vertex then has exactly $N$ neighbours, the adjacency matrix has constant row and column sums, and the local spin-flip operators, being Pauli matrices acting on distinct tensor factors, commute with one another (Figure 2).

This maximal symmetry makes the model exactly solvable. The adjacency operator is diagonalised in the joint eigenbasis of the local operators, its eigenvalues are sums of $N$ independent Rademacher variables, and the classical CLT yields a Gaussian density of states, with the binomial degeneracies controlling the finite-size corrections (Figure 3). The model thereby is a pedagogical warm-up, and the following sections establish to what extent of this Gaussian behaviour survives once commutativity and the constancy of the row sums are abandoned. The elementary derivation, including the degeneracy formula and the uniform error term, is reproduced in Appendix B.

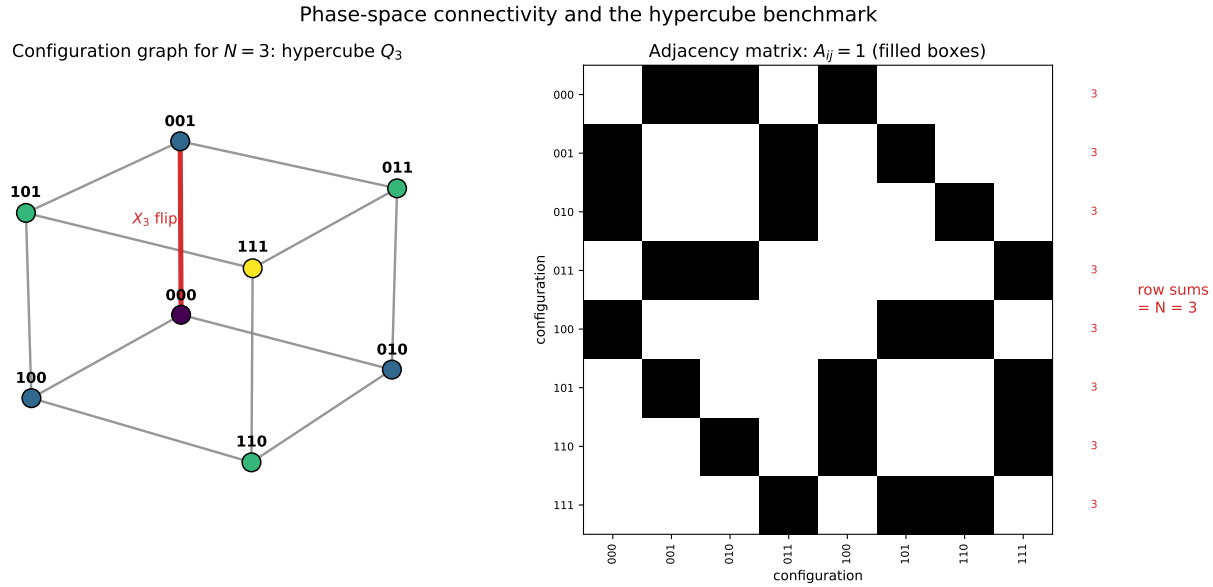



**Figure 2.** The $N = 3$ hypercube configuration graph and its adjacency matrix. Every vertex has degree $N = 3$, corresponding to constant row sums.

## 2 From commuting to non-commuting local operators

The hypercube model owes its solvability to the local commutative operators and vertices of the configuration graph having the same degree. Nevertheless, neither feature is essential to the Gaussian limit. The limit does require locality in the sense that correlations of the system decay fast enough for the global operator to behave like a sum of weakly dependent random variables. To express this without reference to a particular lattice geometry, we work in the operator algebra of the configuration space (Figure 4).

Let $\Lambda = \mathbb{Z}_{L_1} \times \cdots \times \mathbb{Z}_{L_d}$ be a periodic lattice with $N = L_1 \cdots L_d$ sites and $\mathcal{H}$ the associated $2^N$-dimensional Hilbert space. Fix a Hermitian operator $X_0$ and let $X_i$ be its translates under the commuting shift operators of the lattice, so that the family $\{X_i\}$ is translation covariant by

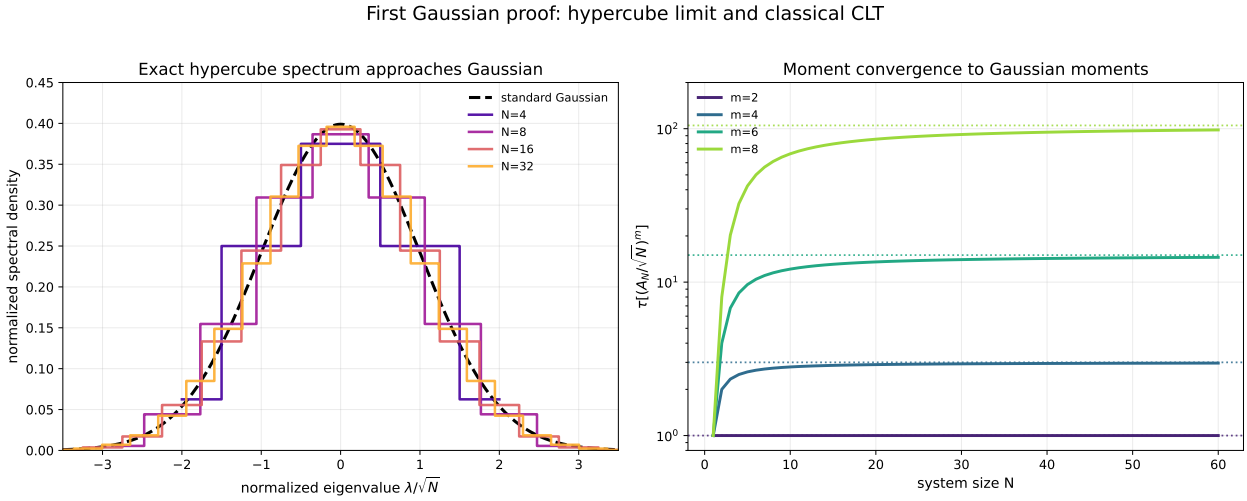



**Figure 3.** Exact hypercube spectrum for $N = 4, 8, 16, 32$ approaching the standard Gaussian, and convergence of even moments to $(m-1)!!$.

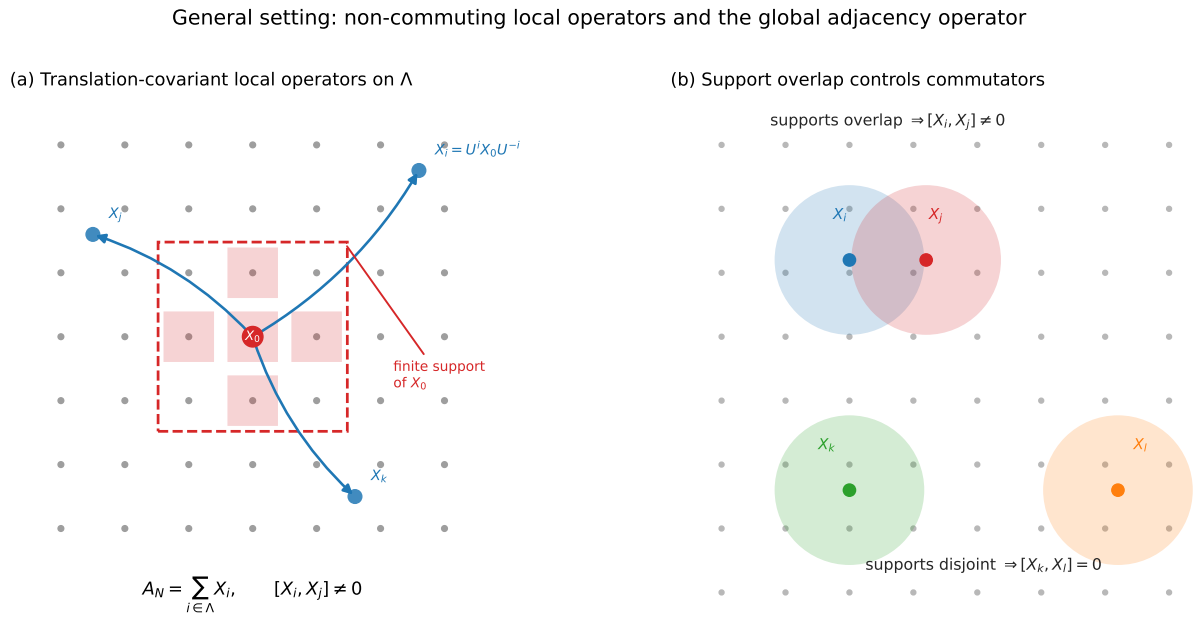



**Figure 4.** General setting. (a) Translation-covariant local operators $X_i = U^i X_0 U^{-i}$ on a periodic lattice, with finite support around the origin. (b) The commutator $[X_i, X_j]$ is controlled by the overlap of supports: overlapping supports give non-commuting operators, while disjoint supports commute.

construction. The global adjacency operator is $A_N = \sum_{i\in\Lambda} X_i$. Since the $X_i$ need not commute, $A_N$ cannot be diagonalised explicitly, and its spectral density is probed instead through the normalised traces $\tau\big((A_N/\sqrt{N})^m\big)$ of powers of the normalised operator, where $\tau$ denotes the trace divided by $2^N$.

The hypothesis that carries the physics is quasi-locality, namely the exponential decay of nested commutators in the diameter of the supports involved:

$$\big\| \big[X_{i_1}, [X_{i_2}, \cdots [X_{i_{k-1}}, X_{i_k}] \cdots]\big] \big\| \leq C'_k \, e^{-\mu' \operatorname{diam}(i_1,\ldots,i_k)}, \qquad k \geq 2, \tag{A4}$$

with constants $C'_k > 0$ and $\mu' > 0$ independent of $N$. This is the operator-algebraic form of the Lieb–Robinson bound [9] which states that information propagates at a finite speed and $X_i$ and $X_j$ act almost independently when their supports are far apart. For strictly local $X_0$ the commutator on the left vanishes identically whenever the supports are disjoint, and (A4) holds trivially with constants uniform in the lattice size. This is why the models discussed in Section 5 fall inside the framework without further input, and it is the property we regard as the core of the theory; for finite-range operators it implies the clustering of cumulants used below (Appendix C).

The hypotheses used in the proof are stated in a slightly different form. In addition to Hermitian, translation covariance and the zero-mean condition, they comprise the exponential clustering of cumulated trace and its classical counterpart for the row-sum functions; these are listed in full in Appendix C, and the discussion there explains how they are related to (A4) and why they are sufficient rather than necessary. The cumulant expansion and the combinatorics of the moment method are developed in Appendix D. Under these hypotheses the limit takes the following form.

**Theorem 2.1** (Gaussian limit for non-commuting sums)**.** *Under assumptions (A1)–(A3), the ESD of*

*$\frac{A_N}{\sqrt{N}}$ converges weakly, as $N \to \infty$ with $L_1, \ldots, L_d \to \infty$, to the centred Gaussian law with variance*

$$\sigma^2 := \lim_{N\to\infty} \sum_{j\in\Lambda} \kappa_2(X_0, X_j) = \lim_{N\to\infty} \sum_{j\in\Lambda} \tau(X_0 X_j), \tag{2.1}$$

*which is finite and non-negative. More precisely, for every integer $m \geq 0$,*

$$\lim_{N\to\infty} \tau\left[\left(\frac{A_N}{\sqrt{N}}\right)^m\right] = \begin{cases} (m-1)!!\,\sigma^m, & m \text{ even}, \\ 0, & m \text{ odd}. \end{cases}$$

*Consequently,*

$$\boxed{\rho_{\frac{A_N}{\sqrt{N}}}(\lambda) \xrightarrow[N\to\infty]{} \mathcal{N}(0, \sigma^2)} \tag{2.2}$$

*in the weak topology, provided $\sigma^2 > 0$.*

The proof of Theorem 2.1 proceeds by examining the moments of $A_N/\sqrt{N}$, which generalises the counting argument of Han et al. [7], where moments were obtained by enumerating legitimate loops in the configuration graph. Here the Gaussian moment sequence $(2n-1)!!\,\sigma^{2n}$ arises because the dominant contributions come from pairings of the $X_i$'s, each pairing corresponding to acting with $X_j$ on a vertex. Since $A_N$ is traceless by assumption (A2), the limiting distribution is centred at zero. The variance $\sigma^2$, defined in (2.1), is non-trivial and encodes the model-specific correlation strength. In the following sections we will examine the robustness of this Gaussian limit under additional symmetries and under renormalisation.

# 3 Conserved charges, sector decomposition and connectivity

Following from the construction above, now we apply a conservation law that fragments the spectrum into blocks, and the resulting mixture of block spectra need not be Gaussian even when each block is. To begin with, let $Q$ be a Hermitian operator commuting with $A_N$, in analogy with the particle-number or magnetisation operator of a lattice model. Then $\mathcal{H}$ splits into the eigenspaces $\mathcal{H}_q$ of $Q$, each of which is invariant under $A_N$; the adjacency operator is therefore block diagonal, and the full empirical spectral density is the mixture of the sector densities weighted by $p_q = \dim \mathcal{H}_q / 2^N$ (Figure 5).

Each sector inherits the mechanism of Section 2, where the exponential decay of clustering within a sector with constants uniform in $q$ and $N$ yields Gaussian (conditional) spectral density in that sector. The full spectrum, however, is a mixture of Gaussians and is not Gaussian in general. It collapses to a single Gaussian exactly when the sector variances $\sigma_q^2$ concentrate around their mean (Figure 6). This is a property of the model rather than an artefact of the proof, as the remarks in Appendix E show. A mixture of centred Gaussians is itself Gaussian only if the variance is almost surely constant, so that any non-trivial dependence $q \mapsto \sigma_q^2$ produces a non-Gaussian limit in the exact sense. The sector-wise hypotheses are stated in Appendix E.

**Theorem 3.1** (Gaussian limit under a conservation law)**.** *Suppose that (A1) and (C0)–(C2) hold.*

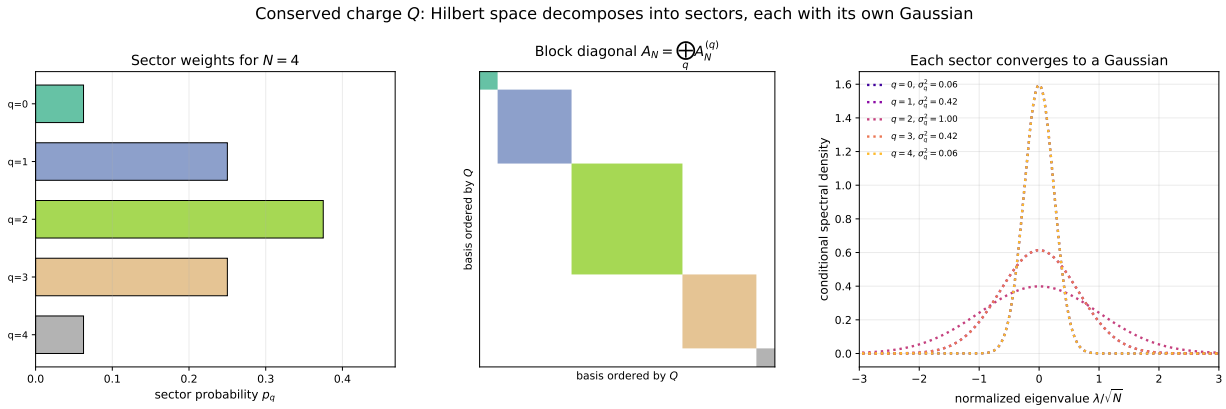



**Figure 5.** Conserved charge and sector decomposition. For a small system with $N = 4$ sites and conserved total particle number $Q = \sum_i n_i$, the Hilbert space splits into sectors $\mathcal{H} = \bigoplus_q \mathcal{H}_q$ with weights $p_q = \dim \mathcal{H}_q / 2^N$ (left). Since $[Q, A_N] = 0$, the adjacency operator is block diagonal, $A_N = \bigoplus_q A_N^{(q)}$ (middle). Within each sector, the conditional spectral density converges to a centred Gaussian with sector variance $\sigma_q^2$ (right).

*For each sector $q$ with $p_q > 0$, define the (finite) sector variance*

$$\sigma_q^2 := \sum_{j \in \Lambda} \kappa_2^{(q)}(X_0, X_j),$$

*Then, for every fixed $q$, the conditional empirical spectral density of $A_N / \sqrt{N}$ restricted to $\mathcal{H}_q$ converges weakly to the centred Gaussian law:*

$$\rho_{\frac{A_N}{\sqrt{N}} \,|\, Q=q}(\lambda) \xrightarrow[N\to\infty]{} \mathcal{N}(0, \sigma_q^2)$$

*If, in addition, the concentration condition (C3) holds, then the unconditional empirical spectral density of $A_N / \sqrt{N}$ converges weakly to*

$$\rho_{\frac{A_N}{\sqrt{N}}}(\lambda) \xrightarrow[N\to\infty]{} \mathcal{N}(0, \sigma^2),$$

*In this case the result reduces exactly to that of Theorem 2.1 and $Q$ is said to be trivial.*

In order to understand the geometry of the configuration network of the phase space, we need to focus on the degree (also called connectivity) of each vertex, which is defined as the number of edges incident to that vertex. For the symmetric adjacency matrix $A_N$, the degree of a configuration vertex is precisely its row sum (equivalently, column sum). In the hypercube model, every vertex has the same degree $N$, which is the consequence of the completely commutative operators $X_j$. For general interacting systems, the row sums become site-dependent random variables, and their correlation are controlled by the classical cumulant decay in assumption (A5). With further restrictions in the form of the Lieb-Robinson bounds on the row-sum functions, one can deduce the asymptotic distribution of the normalised row sums as $N \to \infty$. The following proposition formalises this result.

The result is illustrated in Figure 7, and the proof is a transcription of the proofs of Theorems 2.1 and 3.1, given in Appendix E.4.

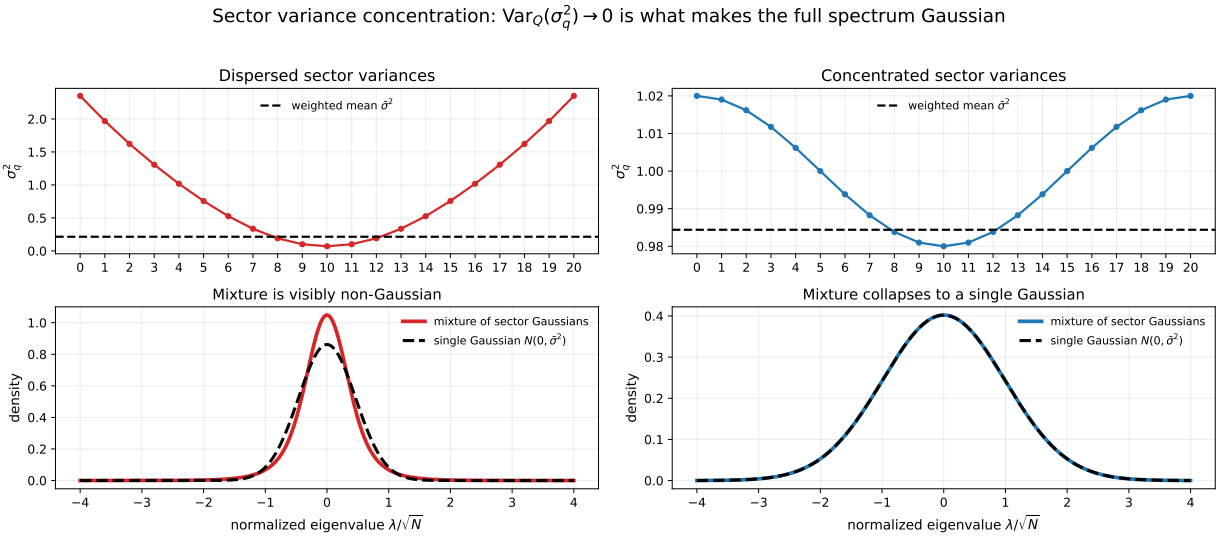



**Figure 6.** Sector variance concentration. The full spectral density is the mixture $\rho_{A_N}(\lambda) = \sum_q p_q \rho_{A_N^{(q)}}(\lambda)$. Left column: when the sector variances $\sigma_q^2$ are dispersed, the mixture remains visibly non-Gaussian even though each sector is asymptotically Gaussian. Right column: when $\sigma_q^2$ concentrates around its weighted mean, the mixture collapses to a single Gaussian. This illustrates why the concentration condition $\mathrm{Var}_Q(\sigma_q^2) \to 0$ is not merely sufficient but forced by the requirement that the full limiting distribution be Gaussian.

**Proposition 3.1** (Gaussian connectivity fluctuations)**.** *Assume (A1)-(A5), then the empirical distribution of*

$$\frac{1}{\sqrt{N}} \sum_{i \in \Lambda} T_i(x)$$

*with respect to the uniform measure on the computational basis converges weakly, as $N \to \infty$, to a centred Gaussian law with variance*

$$\sigma_{\mathrm{row}}^2 := \lim_{N \to \infty} \sum_{j \in \Lambda} \mathbb{E}_x[T_0(x) T_j(x)].$$

*If, in addition, a conserved charge $Q$ is present and the sector assumptions (C0)-(C2)&(C4) hold, then the same conclusion holds in each sector after conditioning, and the global distribution is again Gaussian with variance $\mathbb{E}_Q[\sigma_{q,\mathrm{row}}^2]$, provided the sector variances concentrate as in (C3).*

Conservation laws in real systems are rarely exact, and it is natural to ask how much of the above survives when $Q$ commutes with $A_N$ only approximately. In fact, the sector decomposition survives to leading order, where the off-diagonal part of $A_N$ is suppressed by the commutator, the Gaussian limit persists, and the only trace of the violation is a possible shift of the mean (Figure 8). The precise statement, including the size of the shift and the class of models in which the mechanism fails, is given in Appendix G.

# 4 Decimation and self-similarity

The Gaussian limit is not only robust but also stable under coarse graining, in particular, the block-spin decimation of the renormalisation group [15, 5] acts here on the space of local opera-

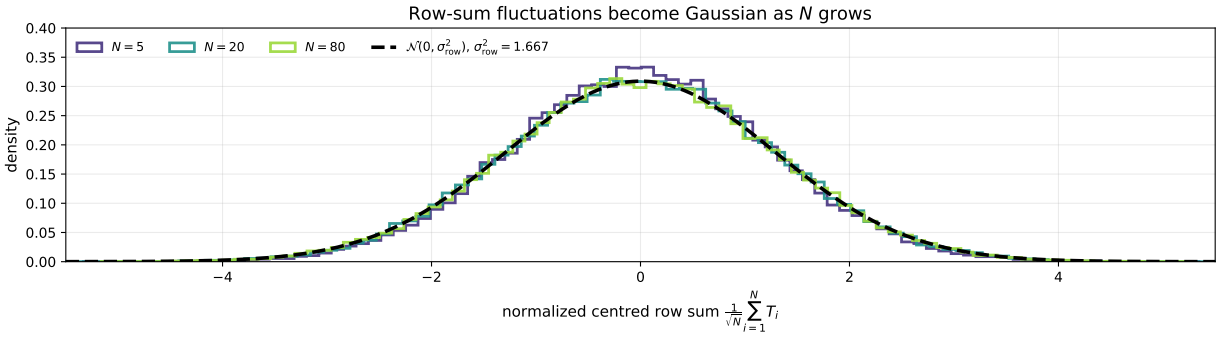



**Figure 7.** Connectivity fluctuations in a toy model. Each local centred row-sum contribution $T_i$ is modelled as a stationary AR(1) sequence with unit marginal variance and correlation $\rho = 0.25$, so that distant contributions are weakly dependent. For this toy model the limiting row-sum variance is $\sigma^2_{\mathrm{row}} = (1+\rho)/(1-\rho) = 5/3 \approx 1.667$. The normalized total row sum $N^{-1/2}\sum_{i=1}^{N} T_i$ is sampled with 60,000 realizations and histogrammed with 90 bins for $N = 5, 20, 80$. As $N$ increases, the empirical distribution approaches $\mathcal{N}(0, \sigma^2_{\mathrm{row}})$ (black dashed line), in agreement with Proposition 3.1 on Gaussian connectivity fluctuations.

tor families rather than on scalar couplings, yet it leaves the Gaussian fixed point intact. When partition the lattice into blocks of linear size $\ell$ and replace each block by the sum of the local operators it contains, we find that the resulting block operators form an approximately translational invariant family on the coarser lattice (Figure 9). Because a block sum is a finite sum, the exponential clustering of the cumulants is preserved with rescaled constants, so that the coarse-grained system falls back into the framework of Section 2 and obeys the same limiting law, up to the deterministic rescaling of the variance which the change of scale entails.

The map from the original family of local operators to the block family is thus a renormalisation group (RG) transformation with a Gaussian fixed point. Iterating it leaves the limiting density

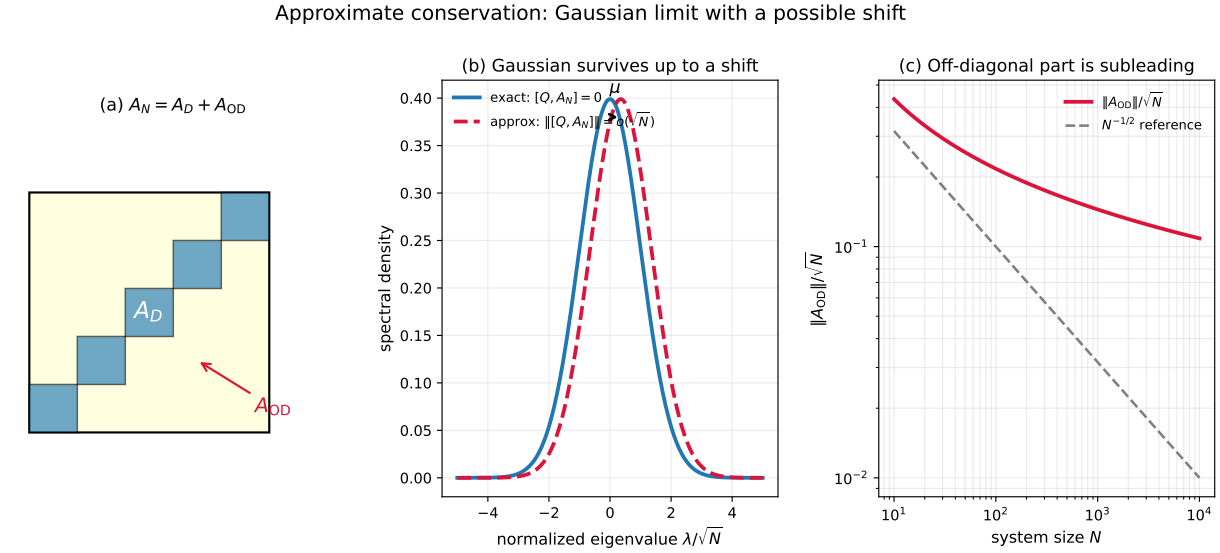



**Figure 8.** Approximate conservation and spectral shift. (a) The global operator decomposes as $A_N = A_D + A_{\mathrm{OD}}$, where $A_D$ is block diagonal with respect to the eigenspaces of $Q$, and $A_{\mathrm{OD}}$ is the off-diagonal part. (b) If the commutator is small in the local sense $\|[Q, A_N]\| = o(\sqrt{N})$, the Gaussian limit survives up to a possible shift $\mu = \lim_{N\to\infty} \tau(A_{\mathrm{OD}})/\sqrt{N}$. (c) The off-diagonal part is subleading: $\|A_{\mathrm{OD}}\|/\sqrt{N} \to 0$ as $N \to \infty$.

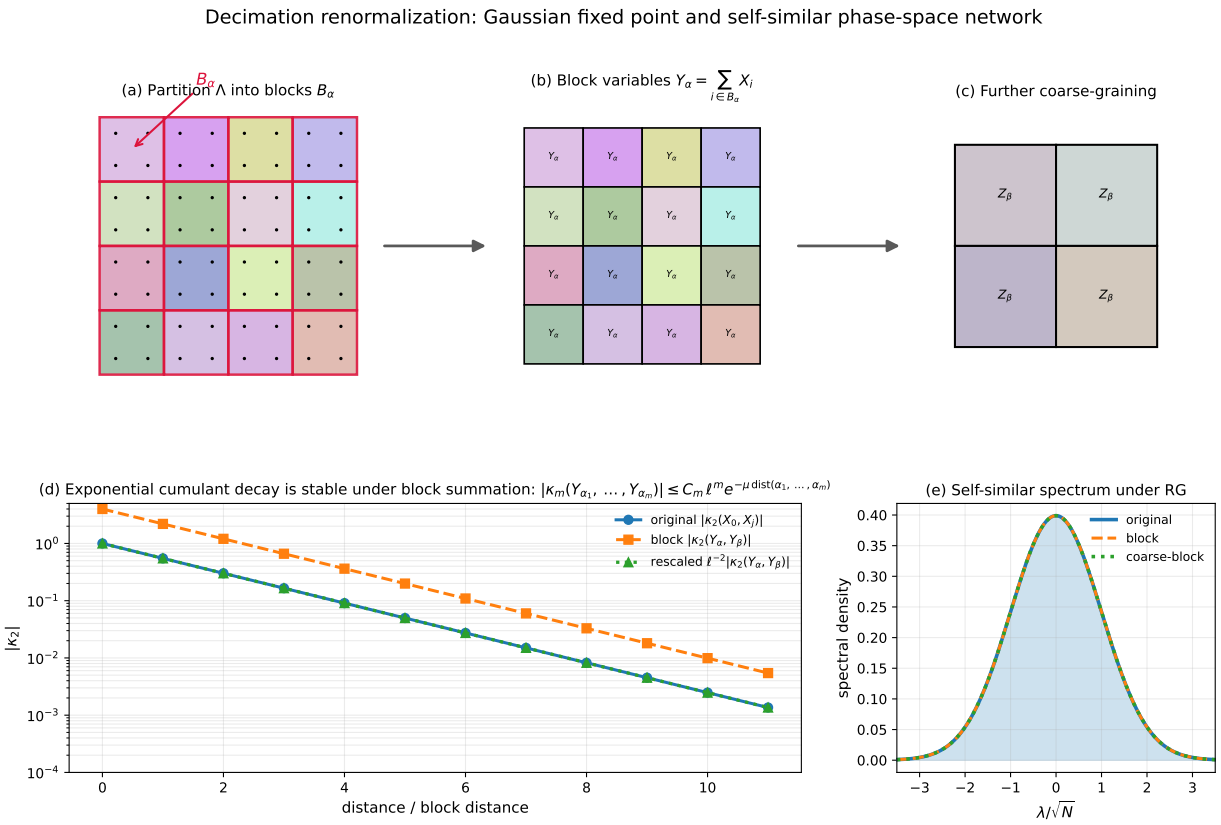



**Figure 9.** Decimation renormalization and the Gaussian fixed point. (a) The lattice $\Lambda$ is partitioned into blocks $B_\alpha$ of linear size $\ell$. (b) The block operators $Y_\alpha = \sum_{i \in B_\alpha} X_i$ form an approximately translation-invariant family on the coarse-grained lattice. (c) Further coarse-graining produces the next-level block variables $Z_\beta$. (d) The exponential decay of cumulants is preserved under block summation, with rescaled constants $|\kappa_m(Y_{\alpha_1}, \ldots, Y_{\alpha_m})| \le C_m\, \ell^m e^{-\mu\, \mathrm{dist}(\alpha_1, \ldots, \alpha_m)}$. (e) Consequently, the normalized spectral density of $A_N / \sqrt{N}$ is invariant under the decimation map, yielding a self-similar Gaussian fixed point.

of states unchanged, and the network is asymptotically self-similar. The transformation is not exact at finite $\ell$, since the non-commutativity of the local operators generates corrections from nested commutators; these corrections are controlled by (A4) and are exponentially small in the separation of the blocks, so that the map is an asymptotic rather than an exact symmetry. The estimates, together with the corresponding statement for the row sums, are collected in Appendix F.

It is worth to notice that, while the details of the highly complex network of configurations is an extremely challenging question, we have the capability to detect (asymptotic) self-similar structures upon RG decimation transformation. In other words, by coarse-graining the physical space and apply decimation, the distributions we stress in this essay stays invariant asymptotically upon rescaling, and we deduce that the network is therefore self-similar. Han et al has investigated examples with such phenomena and named this self similarity as "fractal network". In next section we shall discuss the typical examples they proposed and simulated, with brief discussion on verify the LR bound.

# 5 Discussion of Examples

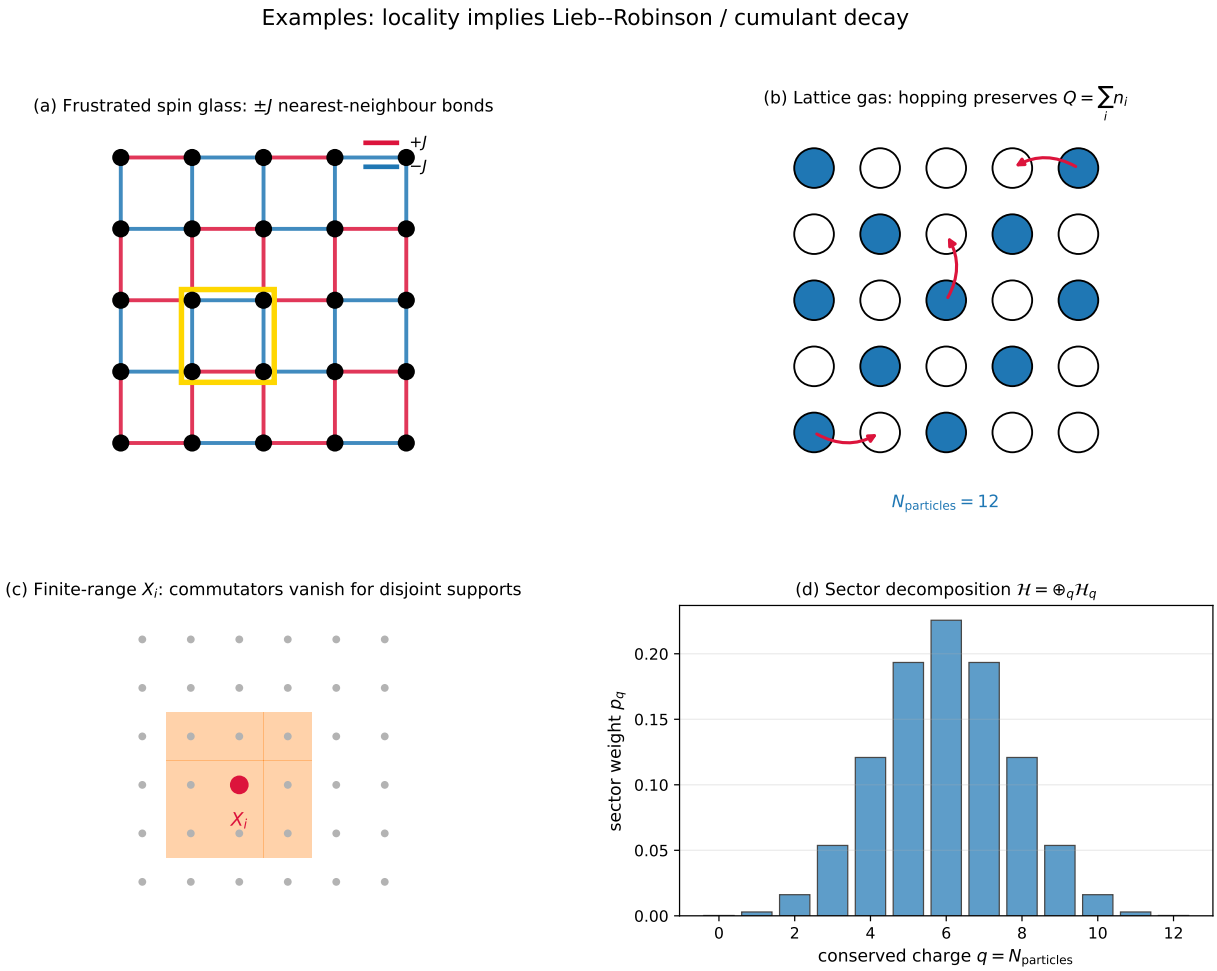



**Figure 10.** Two concrete models satisfying the general hypotheses. (a) Frustrated spin glass with $\pm J$ nearest-neighbour bonds; the highlighted plaquette is frustrated. (b) Lattice gas with conserved particle number; hopping processes exchange occupied and empty sites and preserve $Q = \sum_i n_i$. (c) Finite-range local operators $X_i$ have bounded support, so multiple commutators vanish identically for disjoint supports, and the Lieb–Robinson/cumulant decay assumptions hold trivially. (d) In the lattice gas, the conserved charge induces a sector decomposition $\mathcal{H} = \oplus_q \mathcal{H}_q$; each sector inherits the Gaussian limit, and the sector variances concentrate in the thermodynamic limit.

We now illustrate how the general conditions developed in the preceding sections are satisfied in two canonical models, i.e. the frustrated spin glass and the lattice gas. Both have been studied numerically for small system sizes in the work of Han et al. (2015), where the emergence of Gaussian spectral densities was observed as $N$ increases. Our aim here is not to perform new computations, but rather to indicate why these models naturally fall within the framework of our assumptions, and to highlight the sense in which the verification is essentially trivial once the local structure is identified.

Consider first the frustrated spin glass, whose phase-space network was mapped and analysed in [6, 4]. The Hamiltonian consists only of nearest-neighbour interactions on a lattice, with frustration arising from competing couplings. Since all interactions are strictly local and of

finite range, the associated local operators $X_i$ have finite support. Consequently, the multiple commutators in assumption (A4) vanish identically for disjoint supports, and the exponential decay of cumulated traces in (A3) follows immediately from the product structure of the maximally mixed state. Likewise, the classical cumulant decay in (A5) is automatic because the row-sum functions are built from these same local terms. Thus, apart from the trivial shift of the mean (which can be removed by centring), the spin glass satisfies all assumptions of Theorem 2.1 and Theorem 3.1 without any further input.

For the lattice gas model with a conserved particle number, whose phase-space networks were examined in [7, 11], the situation is analogous. The conserved charge $Q$ is the total particle number operator, and the local terms $X_i$ describe hopping processes that exchange occupied and empty sites on neighbouring lattice sites, where each $X_i$ acts only on a pair of adjacent vertices. This dynamics is manifestly local, hence the sector decomposition induced by $Q$ inherits the same finite support property, and the sector cumulants in (C2) as well as the classical sector cumulants in (C4) decay exponentially for the same reason. Thus the lattice gas with fixed number of particles satisfies all the necessary hypotheses (except for (C3)) of Theorem 3.1, and the global Gaussian limit follows.

In both examples, the clustering assumptions reduces to the finite range interactions. There is therefore no need to invoke more elaborate estimates since the Lieb–Robinson bounds are automatically satisfied with exponential tails, and the cumulant decay constants are uniform in the system size. This explains why the Gaussian universality observed numerically is robust and does not depend on fine details of the model.

Finally, we note that the discussion above has been phrased for a simple cubic lattice for concreteness. For closed-packed structures such as the face-centred cubic (FCC) lattice, one may regard the lattice as a motif convoluted with a simple cubic lattice; the locality and translation covariance properties remain unchanged. For hexagonal close-packed (HCP) structures in three or higher dimensions, the generator of translations is slightly more involved, but the generalisation is straightforward and does not alter the validity of the assumptions. The key requirement of finite range interactions is unaffected by the underlying Bravais lattice.

# 6 Conclusion

In this work, we have formulated a general operator-algebraic framework for the Gaussian universality of phase-space connectivity in lattice-based systems. Under assumptions of translation covariance and exponential decay of cumulated traces, we proved that the ESD of the global adjacency operator converges weakly to a centred Gaussian law. When a conserved charge $Q$ commutes with the global operator, we showed that each charge sector admits a Gaussian limit. We also established the stability of the Gaussian fixed point under decimation-type coarse graining and verified that strictly local models, such as frustrated spin glasses and lattice gases with conserved particle number, fall naturally within the framework.

Several fundamental limitations nevertheless remain before these results can be developed into a practical statistical physics criterion. First, the phase-space network is exponentially large. For small systems, finite-size and boundary effects may obscure the asymptotic spectral behaviour; for large systems, complete enumeration of configurations is computationally prohibitive. It is therefore necessary to devise sampling-based or otherwise scalable methods that can test Gaussianity without diagonalising the full adjacency operator. Second, our analysis con-

cerns the connectivity of configurations only within fixed global eigenvalues of the Hamiltonian or of another conserved charge. It does not by itself predict which configurations are statistically weighted or dynamically observed. Near a phase transition, in the presence of long-range interactions, or when the limiting law is shifted or deformed, such a sector-by-sector description falls outside the scope of the present assumptions. Third, the framework still lacks a systematic thermodynamic treatment. In a fixed-volume setting, the free energy cannot be reduced to a simple configurational entropy, and a naive microcanonical formula such as $S(E) = k \ln \Omega(E)$ becomes delicate near phase transitions. Consequently, we cannot yet predict the associated thermodynamic behaviour from the phase-space connectivity alone.

In summary, we have provided a general proof that extends the previously observed Gaussian spectral phenomenon from specific lattice models to a broad class of local, translation-covariant systems. Whether this approach can eventually be turned into a reliable statistical-physics diagnostic remains an open question. Much foundational work is still required, and the general non-Gaussian case remains largely unexplored.

# A List of Symbols

## A.1 Mathematics symbols

$\mathbb{Z}$ the ring of integers

$\mathbb{Z}_L$ the cyclic group of integers modulo $L$ (used for lattice directions)

$\mathbb{C}$ the field of complex numbers

$\mathbb{E}(S)$ the probabilistic expectation of $S$

$\mathcal{N}(\mu, \sigma^2)$ the Gaussian distribution of mean $\mu$ and variance $\sigma^2$

$\mathcal{H}$ the Hilbert space of a system

$\mathcal{B}(\mathcal{H})$ the space of bounded linear operators on $\mathcal{H}$

$\|\cdot\|$ the operator norm on $\mathcal{B}(\mathcal{H})$

$\oplus$ the direct sum of Hilbert spaces or operators

$\otimes$ the tensor product of Hilbert spaces or operators

$:=$ the left hand side is defined to be the right hand side

## A.2 Physical symbols

$N$ total number of lattice sites (bits)

$A$ adjacency matrix of the configuration graph (as a global operator)

$X_j$ local operator acting on site $j$

$U, V$ unitary shift operators on the two-dimensional torus

$U_k$ unitary shift operator along lattice direction $k$

$\kappa_m$ $m$-th order tracial cumulant

$\tau$ normalized trace on $\mathcal{B}(\mathcal{H})$

$\Lambda$ finite periodic lattice

$S_i$ row-sum function associated with site $i$

$T_i$ centred row-sum function $T_i = S_i - \mathbb{E}[S_i]$

$\lambda$ eigenvalue of the adjacency matrix

$\rho_A$ empirical spectral density of $A$

$\sigma^2$ limiting variance of the Gaussian law

$Q$ conserved charge operator

$q$ eigenvalue (as a sector label) of $Q$

# B The exactly solvable hypercube model

This appendix reproduces the derivation summarised in Section 1. The figures of that section are not repeated here.

To establish a systematic framework for understanding the universality observed in the spectral properties of phase-space connectivity matrices, it is natural to begin with a model with almost the highest possible symmetry. Among all lattice-gas configurations on a finite torus, the graph whose vertices represent all $2^N$ occupation-number configurations and whose edges connect configurations differing by a single spin flip forms a hypercube. This hypercube geometry is distinguished not only by its exact solvability but also by the fact that every vertex has precisely $N$ neighbours. Consequently, the adjacency matrix of this configuration graph has constant row sum and column sum, a property that is intimately related to the commutativity of the local spin-flip operators $X_j$.

## B.1 Model setup on a 2D torus

Consider a lattice gas system defined on a two-dimensional periodic lattice (torus)

$$\Lambda = \mathbb{Z}_{L_1} \times \mathbb{Z}_{L_2}, \qquad N := |\Lambda| = L_1 L_2,$$

where $N$ denotes the total number of classical bits (representing spins or occupied sites). The full configuration (Hilbert) space is

$$\mathcal{H} = \bigotimes_{(i,j)\in\Lambda} \mathbb{C}^2, \qquad \dim \mathcal{H} = 2^N.$$

To incorporate the two independent directions of translational symmetry, we introduce two unitary shift operators $U$ and $V$ acting on $\mathcal{H}$:

$$U\,|s_{i,j}\rangle = |s_{i-1,j}\rangle, \qquad V\,|s_{i,j}\rangle = |s_{i,j-1}\rangle,$$

where $s_{i,j} \in \{0,1\}$ denotes the occupation number at vertex $(i,j)$. These shifts satisfy the exact periodic boundary conditions

$$U^{L_1} = I, \qquad V^{L_2} = I, \qquad [U,V] = 0.$$

Let the local interaction operator at the origin be the Pauli matrix $\sigma_x$ acting solely on that site:

$$X_{0,0} := \sigma_x^{(0,0)} \otimes I_{\Lambda\setminus\{(0,0)\}}.$$

Using the translational symmetry, we define the interaction at an arbitrary vertex $(i,j)$ as

$$X_{i,j} := U^i V^j\, X_{0,0}\, V^{-j} U^{-i}, \qquad (i,j) \in \Lambda.$$

Since the shifts merely permute the tensor factors, this operator is precisely the local $\sigma_x$ acting on vertex $(i,j)$. The adjacency matrix is defined as the sum over all local operators:

$$A := \sum_{(i,j)\in\Lambda} X_{i,j}. \tag{B.1}$$

## B.2 Exact diagonalization: the hypercube limit

We first consider the simplest case where every vertex is completely independent. Since each $X_{i,j}$ acts non-trivially on a distinct tensor factor, they all commute:

$$[X_{i,j}, X_{i',j'}] = 0, \qquad \forall (i,j),(i',j') \in \Lambda.$$

Thus, $A$ is exactly diagonalized in the joint eigenbasis of all $X_{i,j}$. Let $X_{i,j}|\{\sigma\}\rangle = \sigma_{i,j}|\{\sigma\}\rangle, \sigma_{i,j} \in \{\pm 1\}$. Then the eigenvalue corresponding to the configuration $\{\sigma\} \in \{\pm 1\}^{\Lambda}$ is

$$\lambda(\{\sigma\}) = \sum_{(i,j)\in\Lambda} \sigma_{i,j} = \sum_{n=1}^{N} \sigma_n,$$

where we relabelled the $N$ sites by a single index $n$ for notational convenience. The degeneracy of an eigenvalue $\lambda = N - 2k$ (with $k$ being the number of $-1$'s) is given by the binomial coefficient

$$\deg(\lambda) = \binom{N}{k}, \qquad k = \frac{N-\lambda}{2}. \tag{B.2}$$

The empirical spectral density is the distribution of the sum of $N$ Rademacher independent variables:

$$\rho_A(\lambda) = \frac{1}{2^N} \sum_{\{\sigma\}} \delta\left(\lambda - \sum_{n=1}^{N} \sigma_n\right).$$

By the classical CLT (CLT), as $N \to \infty$ (with $L_1, L_2 \to \infty$),

$$\boxed{\rho_A(\lambda) = \frac{1}{\sqrt{2\pi N}} \exp\left(-\frac{\lambda^2}{2N}\right) + o(1),} \tag{B.3}$$

where the error term is uniform in $\lambda$. This is the rigorous Gaussian limit for the hypercube graph.

In this extremally symmetric setting, all local operators $X_j$ commute with one another, so the global adjacency operator $A$ is diagonalised in a common eigenbasis and its spectral density is exactly that of a sum of $N$ independent Rademacher variables. The CLT then immediately yields a Gaussian limiting density of states. While this hypercube limit serves as an important reference point, the real challenge and the main focus of this work lies in understanding how the Gaussian universality survives when we perturb away from this highly symmetric structure. The subsequent sections develop a general framework in which the local operators need not commute, and the constancy of the row sums is replaced by the much weaker condition of exponential decay of correlations. The hypercube model thus provides both a pedagogical starting point and a sharp limiting case against which more general results can be tested.

# C The general framework and its hypotheses

Having established the Gaussian limit for the highly symmetric hypercube model, we now turn to a more general framework that captures the universal behaviour of a broader class of lattice systems. Consider a set of bits placed on a high-dimensional periodic lattice, where the interaction strength between two sites is governed by their spatial distance. To encode the locality of interactions, we introduce local operators $X_i$ associated with each lattice site and impose periodic boundary conditions through unitary shift operators. The concept of Lieb–Robinson bounds, borrowed from the theory of quantum many-body systems, provides a natural tool to quantify the spatial decay of correlations. Our central goal is to demonstrate that, under sufficiently strong locality conditions, the ESD of the global adjacency operator $A_N = \sum_i X_i$ again converges to a Gaussian law. To this end, we propose a set of sufficient conditions in terms of exponential decay of trace cumulants to guarantee a Gaussian limit. We emphasize that these conditions are by no means necessary; there may exist weaker assumptions that yield the same conclusion. Conversely, for certain physical models, verifying the Gaussian limit may require imposing even stronger decay properties than those stated here.

## C.1 Generalized local operators

Let the lattice be $\Lambda = \mathbb{Z}_{L_1} \times \cdots \times \mathbb{Z}_{L_d}$ with $d$ independent directions, and set

$$N := |\Lambda| = L_1 L_2 \cdots L_d.$$

Let $\mathcal{H}$ be the associated $2^N$-dimensional Hilbert space. We fix a Hermitian operator $X_0 \in \mathcal{B}(\mathcal{H})$, not necessarily acting only on the origin, and define its translates by

$$X_i := U_1^{i_1} U_2^{i_2} \cdots U_d^{i_d} \, X_0 \, U_d^{-i_d} \cdots U_2^{-i_2} U_1^{-i_1}, \qquad i = (i_1, \ldots, i_d) \in \Lambda,$$

where $U_1, \ldots, U_d$ are unitary shift operators satisfying $U_k^{L_k} = I$ for each $k$ and $[U_k, U_l] = 0$ for all $k, l$. Thus the family $\{X_i\}$ is translation covariant by construction. We define the global operator

$$A_N := \sum_{i \in \Lambda} X_i. \tag{C.1}$$

Because the $X_i$ need not commute, $A_N$ is generally not diagonalizable in a common eigenbasis. Nevertheless, under suitable decay conditions its spectral distribution still converges to a Gaussian law in the thermodynamic limit. Let

$$\tau(\cdot) := \frac{1}{2^N} \operatorname{tr}(\cdot)$$

be the normalized trace on $\mathcal{B}(\mathcal{H})$. We will prove convergence of the empirical spectral measure of $A_N / \sqrt{N}$ by the method of moments.

## C.2 Assumptions

We impose the following conditions.

(A1) *Hermiticity and translation covariance.* Each $X_i$ is Hermitian, $X_i = X_i^\dagger$, and the family is translation covariant:

$$X_i = U_1^{i_1} \cdots U_d^{i_d}\, X_0\, U_d^{-i_d} \cdots U_1^{-i_1}.$$

(A2) *Zero mean.*

$$\tau(X_i) = 0 \qquad \forall\, i \in \Lambda.$$

(A3) *Exponential decay of cumulated traces.* For each ordered $m$-tuple $(i_1, \ldots, i_m)$, define the classical cumulant $\kappa_m(X_{i_1}, \ldots, X_{i_m})$ (see Appendix D.1) recursively by

$$\tau(X_{i_1} \cdots X_{i_m}) = \sum_{\pi \in \Pi_m} \prod_{B \in \pi} \kappa_{|B|}(X_{i_a} : a \in B),$$

where $\Pi_m$ is the set of all partitions of $\{1, \ldots, m\}$. This recursion defines $\kappa_m$ uniquely without commutativity assumptions. We assume that there exist constants $C_m > 0$ and $\mu > 0$, independent of $N$, such that for every $m \geq 2$ and all ordered tuples $(i_1, \ldots, i_m)$,

$$\left|\kappa_m(X_{i_1}, \ldots, X_{i_m})\right| \leq C_m\, e^{-\mu\ \mathrm{diam}(i_1, \ldots, i_m)}.$$

(A4) *Quasi-locality (optional; the inequality is displayed as (A4) in Section 2).* There exist constants $C_k' > 0$ and $\mu' > 0$ such that for all $k \geq 2$,

$$\left\| \, [X_{i_1}, [X_{i_2}, \cdots [X_{i_{k-1}}, X_{i_k}] \cdots ]] \, \right\| \leq C_k'\, e^{-\mu'\ \mathrm{diam}(i_1, \ldots, i_k)}.$$

(A5) *Exponential decay of classical cumulants of centred row-sum functions.* For each site $i \in \Lambda$, define the classical function

$$S_i(x) := \sum_{y \in \{0,1\}^N} \langle x | X_i | y \rangle,$$

and let

$$T_i(x) := S_i(x) - \mathbb{E}_x[S_i],$$

where $\mathbb{E}_x$ denotes the uniform average over the computational basis. Define the classical cumulants $\kappa_m^{\mathrm{cl}}(T_{i_1}, \ldots, T_{i_m})$ with respect to $\mathbb{E}_x$ by the same recursion as in (A3), with the trace $\tau$ replaced by the expectation $\mathbb{E}_x$. We assume that there exist constants $C_m > 0$ and $\mu > 0$, independent of $N$, such that for all $m \geq 2$ and all ordered tuples $(i_1, \ldots, i_m)$,

$$\left|\kappa_m^{\mathrm{cl}}(T_{i_1}, \ldots, T_{i_m})\right| \leq C_m\, e^{-\mu\ \mathrm{diam}(i_1, \ldots, i_m)}.$$

**Remark C.1.** Assumptions (A1) and (A2) concern the basic algebraic and statistical properties of the local operators. It is natural to assume Hermitian $X_i$ since the definition of a spin flip allows the identification of $0$ and $1$ up to a sign, the symmetry of the configuration space supports a Hermitian representation of the local operation. Therefore this in turn ensures that all eigenvalues of the global adjacency operator are real. By construction, the adjacency matrix of the configuration graph has zero diagonal and is symmetric; the operator $X_i$ represents the effective legitimate operation centred at site $i$. When the translational symmetry operations are properly defined, traceless $A_0$ (i.e. $\tau(X_0) = 0$) gives rise to the zero-mean condition (A2), which guarantees that

the average contribution of each site vanishes.

Assumptions (A3) and (A4) are closely related. Assumption (A3) requires that all connected correlations of the local operators with respect to the maximally mixed state decay exponentially with the spatial diameter. This exponential decay ensures that distant operators are effectively independent, so that the sum $A_N$ behaves like a sum of weakly dependent random variables. Assumption (A4) is not directly used in the moment computation, but it provides a sufficient condition based on the decay of multiple commutators. Indeed, for operators generated by finite-depth circuits or by short-time evolution under a local Hamiltonian, Lieb-Robinson bounds imply the required exponential decay of commutators; for strictly local $X_0$ with finite support, commutators vanish identically for disjoint supports, so (A4) holds and (A3) follows. The general mechanism behind this implication is that connected correlations of order higher than two decay exponentially outside the Lieb–Robinson cone, as shown in [1].

The classical cumulant decay in assumption (A5) is the counterpart of (A3) for the row-sum functions $T_i$. While (A3) controls the cumulants of the operator entries, (A5) imposes the same exponential decay on the classical cumulants of the centred row sums. This is precisely the condition that generalises the constant row sum of the hypercube model, where every row sum is exactly $N$ and here the row sums become random variables with exponentially decaying correlations. We will show in Proposition 3.1 that, under (A5), the empirical distribution of the normalised row sums also converges to a Gaussian law, thereby establishing a classical analogue of the main theorem.

The limit theorem itself is stated in Section 2 (Theorem 2.1); its proof occupies Appendix D. The toy model used to calibrate the decay constants of the cumulants, and the convergence of the partial sums of the limiting variance, are shown in Figure 11.

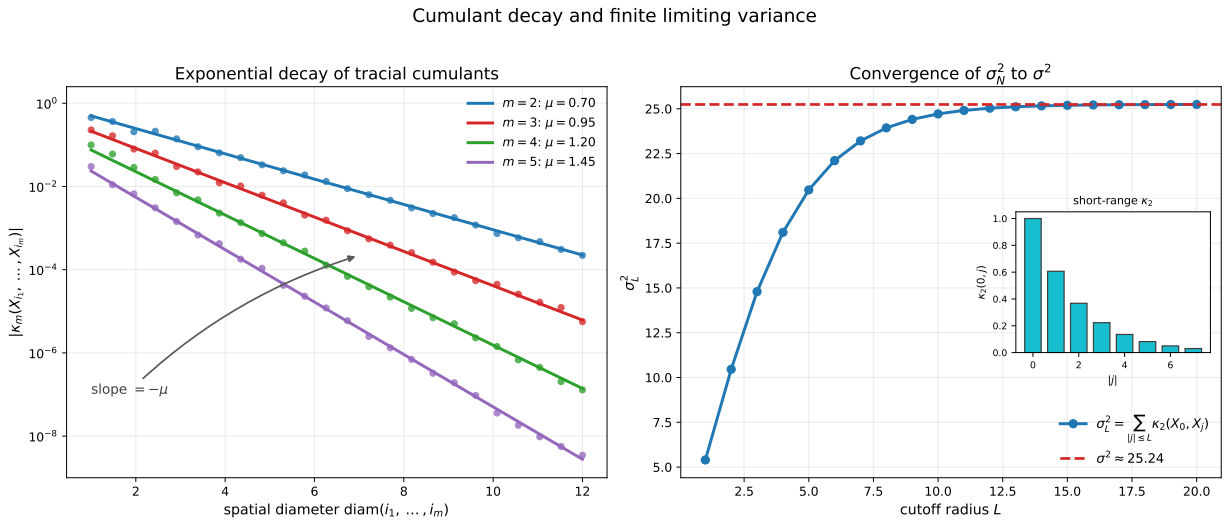



**Figure 11.** Locality input and convergence of the limiting variance, illustrated with a toy model. **Left:** exponential decay of cumulated traces, for $m = 2, 3, 4, 5$. The toy-model parameters are $(C_2, C_3, C_4, C_5) = (1.00, 0.55, 0.25, 0.10)$ and $(\mu_2, \mu_3, \mu_4, \mu_5) = (0.70, 0.95, 1.20, 1.45)$; the spatial diameter is sampled as $d = 1, \ldots, 12$, and multiplicative log-normal noise with standard deviation $0.11$ mimics finite-size fluctuations. **Right:** partial sum $\sigma_L^2 = \sum_{|j| \le L} \kappa_2(X_0, X_j)$ for a short-range second cumulant $\kappa_2(0, j) = Ae^{-|j|/\xi}$ with $A = 1.0$ and correlation length $\xi = 2.0$, computed on a two-dimensional square cutoff of radius $L = 1, \ldots, 20$. The red dashed line is the limiting variance $\sigma^2 = \lim_{L \to \infty} \sigma_L^2$. The inset shows the short-range behaviour of $\kappa_2(0, j)$ for $j = 0, \ldots, 7$. This toy model illustrates assumption (A3) and the finite limiting variance entering Theorem 2.1.

# D The cumulant expansion and the proof of Theorem 2.1

The two subsections below supply, respectively, the cumulant calculus and the proof of the limit theorem. The argument is a direct moment computation rather than an appeal to an abstract limit theorem; the prototype of the phenomenon, namely that exponential mixing forces Gaussian moments, is the classical limit theorem for stationary mixing random fields [3], and the bookkeeping of cumulant orders by which the non-Gaussian terms are discarded is the classical one [8].

## D.1 Explicit cumulant formula and examples

For completeness, we provide the closed-form expression for the cumulated traces used in Assumption (A3). Let $X_1, \ldots, X_m$ be operators on $\mathcal{H}$. The $m$-th order cumulant is defined by the Möbius inversion on the partition lattice:

$$\kappa_m(X_1, \ldots, X_m) = \sum_{\pi \in \Pi_m} (-1)^{|\pi|-1} (|\pi| - 1)! \prod_{B \in \pi} \tau\left( \prod_{a \in B} X_a \right),$$

where $\Pi_m$ is the set of all partitions of $\{1, \ldots, m\}$, $|\pi|$ is the number of blocks in $\pi$, and for each block $B = \{a_1 < \cdots < a_k\}$, the product $\prod_{a \in B} X_a$ is taken in increasing index order. This formula is valid without any commutativity assumption; it is the standard partition-lattice Möbius inversion of the theory of non-commutative cumulants [13, 10]. What the present work adds is the

passage to the Gaussian limit, which requires translation covariance and exponential clustering rather than finite-dimensional matrix asymptotics.

**Example 1 (m=2).** For two operators $X, Y$,

$$\kappa_2(X, Y) = \tau(XY) - \tau(X)\tau(Y),$$

which is the usual connected two-point function.

**Example 2 (m=3).** For three operators $X, Y, Z$,

$$\begin{aligned}\kappa_3(X, Y, Z) = \tau(XYZ) - \tau(X)\tau(YZ) - \tau(Y)\tau(XZ) - \tau(Z)\tau(XY) \\ + 2\,\tau(X)\tau(Y)\tau(Z).\end{aligned}$$

If all operators have zero mean (as in Assumption (A2)), this simplifies to

$$\kappa_3(X, Y, Z) = \tau(XYZ) - \tau(X)\tau(YZ) - \tau(Y)\tau(XZ) - \tau(Z)\tau(XY).$$

## D.2 Proof of Theorem 2.1

*Proof.* We use the method of moments, in the pairing (Wick) form familiar from Wigner's moment method [14, 2]. For each integer $m \geq 1$, write

$$M_m(N) := \tau\left[\left(\frac{A_N}{\sqrt{N}}\right)^m\right] = N^{-m/2} \sum_{i_1,\ldots,i_m \in \Lambda} \tau(X_{i_1} \cdots X_{i_m}).$$

Using the cumulant expansion,

$$\tau(X_{i_1} \cdots X_{i_m}) = \sum_{\pi \in \Pi_m} \prod_{B \in \pi} \kappa_{|B|}(X_{i_a} : a \in B),$$

we obtain

$$M_m(N) = N^{-m/2} \sum_{\pi \in \Pi_m} \prod_{B \in \pi} \left[\sum_{\{i_a : a \in B\}} \kappa_{|B|}(X_{i_a} : a \in B)\right],$$

where the outer sum is over all partitions $\pi$ of $\{1, \ldots, m\}$.

By assumption (A2), $\kappa_1(X_i) = \tau(X_i) = 0$, so no block of size $1$ can occur. Hence every block $B \in \pi$ has size $|B| \geq 2$. Let $b = |\pi|$ be the number of blocks. Then

$$m = \sum_{B \in \pi} |B| \geq 2b,$$

so

$$b \leq \frac{m}{2}.$$

Fix a partition $\pi$ with $b$ blocks. For each block $B$, fix one of its indices, say the smallest element. Because the cumulant decays exponentially in the diameter of the block by (A3), the sum over the remaining indices in $B$ is absolutely convergent and uniformly bounded in $N$. Therefore each block contributes a factor of order $O(N)$ from the single free index. Thus the product over blocks

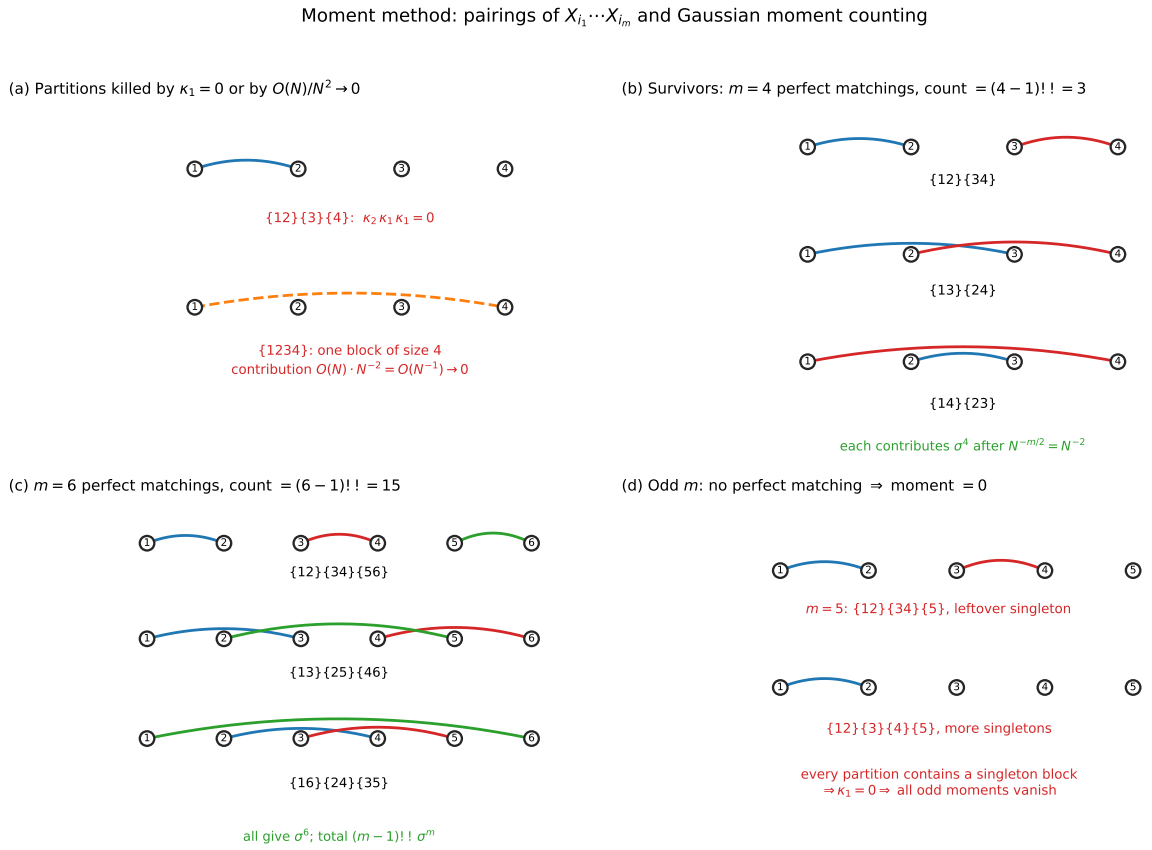



**Figure 12.** Moment method. (a) Partitions containing singleton blocks are killed by $\kappa_1 = 0$, while a single large block contributes $O(N)/N^2 \to 0$. (b) For $m = 4$, the three perfect matchings survive and give $\sigma^4$. (c) For $m = 6$, there are $(6-1)!! = 15$ perfect matchings, each contributing $\sigma^6$. (d) For odd $m$, no perfect matching exists, so every partition contains a singleton and the moment vanishes.

contributes

$$O(N^b).$$

Multiplying by the prefactor $N^{-m/2}$, the contribution of $\pi$ to $M_m(N)$ is

$$O(N^{b-m/2}).$$

If $b < m/2$, this contribution tends to zero as $N \to \infty$. Therefore only partitions with $b = m/2$ can survive. Since $b$ is an integer, this is possible only when $m$ is even, and it forces every block to have size exactly 2. Such partitions are precisely the perfect matchings of $\{1, \ldots, m\}$.

Now let $m$ be even and let $\pi$ be a perfect matching. It consists of $m/2$ pairs. For one pair $(a, b)$, the corresponding sum is

$$\sum_{i,j\in\Lambda} \kappa_2(X_i, X_j).$$

By translation invariance,

$$\kappa_2(X_i, X_j) = \kappa_2(X_0, X_{j-i}),$$

so

$$\sum_{i,j\in\Lambda} \kappa_2(X_i, X_j) = N \sum_{j\in\Lambda} \kappa_2(X_0, X_j) = N\,\sigma_N^2,$$

where we define

$$\sigma_N^2 := \sum_{j\in\Lambda} \kappa_2(X_0, X_j).$$

By (A3) the series is absolutely convergent uniformly in $N$, so

$$\sigma^2 := \lim_{N\to\infty} \sigma_N^2$$

exists and is finite. Moreover,

$$\sigma^2 = \lim_{N\to\infty} \frac{1}{N}\tau(A_N^2) \geq 0,$$

so $\sigma^2$ is non-negative.

Thus each perfect matching contributes

$$N^{m/2}\,(\sigma_N^2)^{m/2}$$

before multiplying by $N^{-m/2}$. After the prefactor, the contribution becomes

$$(\sigma_N^2)^{m/2} \xrightarrow[N\to\infty]{} \sigma^m.$$

The number of perfect matchings of $m$ elements is

$$(m-1)!! = (m-1)(m-3)\cdots 1.$$

Therefore

$$\lim_{N\to\infty} M_m(N) = \begin{cases} (m-1)!!\,\sigma^m, & m \text{ even}, \\ 0, & m \text{ odd}. \end{cases}$$

These are exactly the moments of a centred Gaussian random variable with variance $\sigma^2$. Since the Gaussian distribution is uniquely determined by its moments, the ESD of $A_N/\sqrt{N}$ converges weakly to $\mathcal{N}(0, \sigma^2)$. □

# E Conserved sectors: hypotheses, remarks and proofs

Theorem 3.1 and Proposition 3.1 are stated in Section 3; this appendix records the hypotheses they rest on, discusses their role, and supplies the deferred proofs.

In this appendix we extend the discussion to a common setting in soft-matter physics, where the system is subject to a symmetry or a conservation law. Rather than invoking Noether's theorem, we characterise a conservation law by an operator $Q$ that commutes with the global operator $A_N$, in analogy with the Hamiltonian and number operator in quantum mechanics. Under the same exponential decay conditions, of the Lieb–Robinson type used in Section 2, we obtain a Gaussian distribution for the eigenvalues of the (normalised) operator. However, the precise structure beyond these assumptions in practice often requires careful attention; we defer a detailed discussion of examples and counterexamples to a later section.

## E.1 Conserved quantity and sector decomposition

Let $Q$ representing the desired symmetry be a Hermitian operator on $\mathcal{H}$ such that

$$[Q, A_N] = 0.$$

Write its spectral decomposition, with $P_q$ the orthogonal projection onto the eigenspace

$$Q = \sum_q q\, P_q, \qquad \mathcal{H}_q := P_q \mathcal{H}.$$

Since $A_N$ commutes with $Q$, each $\mathcal{H}_q$ is invariant under $A_N$. We denote by $\mathcal{H}_q := \ker(Q - qI)$ the eigenspace of $Q$ with eigenvalue $q$, and by $P_q$ the orthogonal projection onto $\mathcal{H}_q$. We denote the sector as $Q = q$, then the normalized trace $\tau_q$ and the probability $p_q$ of sector $q$ as:

$$\tau_q(\cdot) := \frac{1}{\dim \mathcal{H}_q} \operatorname{tr}_{\mathcal{H}_q}(\cdot), \qquad p_q := \frac{\dim \mathcal{H}_q}{2^N}.$$

Such decomposition allows us to discuss and verify the condition for a Gaussian spectrum independently in each sector, and conditions for the overall Gaussian spectrum relies on such restriction to be consistent on all sectors of Q.

## E.2 Hypotheses

We impose the following conditions, which are the natural analogues of (A2)–(A3) in each sector, together with a concentration requirement.

(C0) *Translation invariance of the conserved sectors.* We assume that the conserved charge $Q$ commutes with all lattice translation operators:

$$[Q, U_k] = 0 \qquad \forall\, k = 1, \ldots, d.$$

Equivalently, each eigenspace $\mathcal{H}_q$ is invariant under translations. Consequently, the sector trace $\tau_q$ satisfies

$$\tau_q(X_{i+a}) = \tau_q(X_i) \qquad \forall\, a, i \in \Lambda,$$

and, more generally, the sector cumulants are translation invariant:

$$\kappa_m^{(q)}(X_{i_1+a},\dots,X_{i_m+a}) = \kappa_m^{(q)}(X_{i_1},\dots,X_{i_m}) \qquad \forall\, a \in \Lambda.$$

(C1) *Zero mean in typical sectors.* For all $q$ with $p_q > 0$,

$$\tau_q(X_i) = 0.$$

(In practice, a less strict condition is proposed in Appendix G.)

(C2) *Exponential clustering of sector cumulants.* For each $q$ and each ordered $m$-tuple $(i_1,\dots,i_m)$, define the sector cumulants $\kappa_m^{(q)}(X_{i_1},\dots,X_{i_m})$ recursively by the same formula as in (A3), but with $\tau$ replaced by $\tau_q$. We assume that there exist constants $C_m > 0$ and $\mu > 0$, independent of $q$ and $N$, such that for all $m \geq 2$,

$$\left|\kappa_m^{(q)}(X_{i_1},\dots,X_{i_m})\right| \leq C_m\, e^{-\mu\ \mathrm{diam}(i_1,\dots,i_m)}.$$

(C3) *Concentration of the sector variance.* For each $q$, define

$$\sigma_q^2 := \sum_{j\in\Lambda} \kappa_2^{(q)}(X_0, X_j).$$

We assume that $\sigma_q^2$ is finite uniformly in $N$ and that, with respect to the distribution $\{p_q\}$,

$$\mathrm{Var}_Q\left(\sigma_Q^2\right) \xrightarrow[N\to\infty]{} 0.$$

Equivalently, the sector variance is essentially constant on typical sectors.

(C4) *Exponential decay of sector classical cumulants of centred row-sum functions.* For each sector $q$, let

$$T_i^q(x) := S_i(x) - \mathbb{E}_q[S_i],$$

where $\mathbb{E}_q$ is the uniform average over basis states in $\mathcal{H}_q$. Define the classical cumulants $\kappa_m^{\mathrm{cl},q}(T_{i_1}^q,\dots,T_{i_m}^q)$ with respect to $\mathbb{E}_q$ as in (C2), with the operator variables replaced by the functions $T_i^q$. We assume that there exist $C_m > 0$ and $\mu > 0$, independent of $q$ and $N$, such that for all $m \geq 2$,

$$\left|\kappa_m^{\mathrm{cl},q}(T_{i_1}^q,\dots,T_{i_m}^q)\right| \leq C_m\, e^{-\mu\ \mathrm{diam}(i_1,\dots,i_m)}.$$

**Remark E.1.** Assumption (C0) is a natural restriction in our framework. It ensures translational invariant conserved charge and is essential to valid the CLT argument. In the presence of a strong centre or a point source that breaks translation invariance, the basic premise of the CLT summation would be destroyed. Additionally, for system of negligible edge effects, periodic boundary condition is consistent with (C0) and enables the theoretical analysis. In the thermodynamic limit $N \to \infty$, boundary effects are negligible in any case, so (C0) is both physically intuitive and technically convenient.

Assumption (C1) requires the sector trace $\tau_q$ to have zero mean, which is the sector analogue of (A2). While (A2) is global, (C1) imposes the same condition on each conserved sector. In general,

the sector mean may be non-zero even if the global mean vanishes, and the precise relation between the sector means and the global mean is discussed in Appendix G, where we show that a non-zero sector mean can be absorbed by a phase shift in the limiting distribution without altering its Gaussian nature, provided the shift is of order $1/\sqrt{N}$.

The exponential decay of sector cumulants in (C2) is the direct analogue of (A3) within each conserved sector. It is the key input that ensures the independence of distant sites even after conditioning on the value of the conserved charge. Similarly, assumption (C4) is the sector version of (A5) for the classical row-sum functions: it imposes exponential decay on the classical cumulants of the centred row sums in each sector, in complete analogy with the relationship between (A3) and (A5). Thus the pair (C2) and (C4) mirrors the pair (A3) and (A5), respectively, the former controlling the operator cumulants and the latter controlling the classical row-sum cumulants.

**Remark E.2.** Assumption (C3) is not an automatic consequence of a conserved charge. Instead, it addresses the concentration of the sector variance $\sigma_q^2$. As established in the proof of Theorem 3.1, conditions (C0)–(C2) already guarantee that, within each fixed sector $q$, the conditional distribution of $A_N/\sqrt{N}$ converges to a centred Gaussian with variance $\sigma_q^2$. However, the global (unconditional) spectral distribution is a mixture of these sector Gaussians weighted by $p_q$. Such a mixture is not generally Gaussian unless the variances $\sigma_q^2$ are essentially constant across the sectors that carry significant weight. Assumption (C3) ensures precisely the condition of trivial symmetry, because it requires the variance of $\sigma_Q^2$ with respect to the sector distribution vanishes in the thermodynamic limit, and the mixture collapses to a single Gaussian with variance $\sigma^2 = \lim_{N\to\infty} \mathbb{E}_Q[\sigma_Q^2]$.

To see this explicitly, note that the sector decomposition induced by the conserved quantity $Q$ is orthogonal:

$$\mathcal{H} = \bigoplus_q \mathcal{H}_q, \qquad P_q P_{q'} = \delta_{q,q'} P_q.$$

Since $[Q, A_N] = 0$, the adjacency matrix is block diagonal with respect to this decomposition:

$$A_N = \bigoplus_q A_N^{(q)}, \qquad A_N^{(q)} := P_q A_N P_q.$$

Consequently, the full empirical spectral density is the weighted mixture

$$\rho_{A_N}(\lambda) = \sum_q p_q \, \rho_{A_N^{(q)}}(\lambda), \qquad p_q = \frac{\dim \mathcal{H}_q}{2^N}.$$

Thus, even if each sector converges to a Gaussian, the global limit is a Gaussian mixture; the condition (C3) is precisely what forces that mixture to be a single Gaussian.

**Remark E.3.** Suppose now that, as in the theorem, in each sector $q$ the spectral density of $A_N^{(q)}/\sqrt{N}$ converges to a centered Gaussian with variance $\sigma_q^2$. Then the characteristic function of the full normalized adjacency matrix is, in the limit,

$$\Phi(t) = \lim_{N\to\infty} \tau\left(e^{itA_N/\sqrt{N}}\right) = \sum_q p_q \, e^{-t^2\sigma_q^2/2} = \mathbb{E}_Q\left[e^{-t^2\sigma_Q^2/2}\right].$$

If we require the full limiting distribution to be exactly Gaussian with variance $\sigma^2$, then for all $t$,

$$\mathbb{E}_Q\left[e^{-t^2\sigma_Q^2/2}\right] = e^{-t^2\sigma^2/2}.$$

By Jensen's inequality applied to the convex function $x \mapsto e^{-t^2x/2}$ for $t \neq 0$, we have

$$\mathbb{E}_Q\left[e^{-t^2\sigma_Q^2/2}\right] \geq e^{-t^2\,\mathbb{E}_Q[\sigma_Q^2]/2}.$$

Equality holds if and only if $\sigma_Q^2$ is almost surely constant and equal to $\mathbb{E}_Q[\sigma_Q^2]$. Therefore, if the full limit is a single Gaussian, then necessarily

$$\sigma_Q^2 = \sigma^2 \quad \text{almost surely with respect to } \{p_q\}.$$

In other words, the concentration condition (C3) is not merely a sufficient bound, it is forced by the requirement that a mixture of sector-wise Gaussians again be Gaussian. Thus, any attempt to replace (C3) by a more general functional relation $q \mapsto \sigma_q^2$ will, in the exact limit, either violate overall Gaussian or reduce to the constant-variance case. If one allows only approximate Gaussian, then small fluctuations of $\sigma_q^2$ around a typical value produce a scale mixture of Gaussians, which is generally not Gaussian but may still have Gaussian-like tails. A more general analysis could therefore consider such scale mixtures or allow non-zero sector means, but those lie outside the strict Gaussian framework considered here.

## E.3 Proof of Theorem 3.1

*Proof.* Fix a sector $q$. By (C1)–(C2), the assumptions of Theorem 2.1 are satisfied in the Hilbert space $\mathcal{H}_q$ with trace $\tau_q$. Therefore, for every integer $m \geq 0$,

$$\lim_{N\to\infty} \tau_q\left[\left(\frac{A_N}{\sqrt{N}}\right)^m\right] = \begin{cases} (m-1)!!\,\sigma_q^m, & m \text{ even}, \\ 0, & m \text{ odd}. \end{cases}$$

Hence the moments of $\frac{A_N}{\sqrt{N}}$ in sector $q$ converge to those of $\mathcal{N}(0, \sigma_q^2)$.

Now the full characteristic function of $\frac{A_N}{\sqrt{N}}$ is

$$\Phi_N(t) := \tau\left(e^{it\frac{A_N}{\sqrt{N}}}\right) = \sum_q p_q\,\tau_q\left(e^{it\frac{A_N}{\sqrt{N}}}\right).$$

For fixed $t$, the previous moment convergence implies

$$\lim_{N\to\infty} \tau_q\left(e^{it\frac{A_N}{\sqrt{N}}}\right) = e^{-t^2\sigma_q^2/2}$$

uniformly on compact sets of $q$ under (C3). Since $\sigma_q^2$ concentrates around $\sigma^2$, we get

$$\Phi_N(t) \xrightarrow[N\to\infty]{} e^{-t^2\sigma^2/2}.$$

By Lévy's continuity theorem, the spectral distribution converges weakly to $\mathcal{N}(0, \sigma^2)$. □

## E.4 Proof of Proposition 3.1

*Proof.* The proof is identical to that of Theorem 2.1 and Theorem 3.1 after the following replacements:

- replace the non-commuting operators $X_i$ by the classical local functions $T_i$ (or $T_i^q$ in the sector case);
- replace the trace $\tau$ by the uniform expectation $\mathbb{E}_x$ (or $\mathbb{E}_q$);
- replace assumption (A3) by (A5) (or (C2) by (C4) in the sector case).

The cumulant expansion, the counting of partitions, and the pairing dominance argument are unchanged. Therefore the moment convergence and the Gaussian limit follow exactly as in the proof of Theorem 2.1 and Theorem 3.1. □

# F Decimation renormalisation

## F.1 The decimation map

To begin with, we discuss whether the present model admits a decimation-type renormalization mapping analogous to that of the Ising model. The short answer is affirmative in the asymptotic sense, provided the locality and clustering assumptions introduced earlier are satisfied. However, the non-commuting nature of the $X_i$ forces the mapping to be understood at the level of operator-valued cumulants rather than as a scalar recursion. Partition the lattice $\Lambda$ into disjoint blocks $\{B_\alpha\}$ of linear size $\ell$, and define the block operator

$$Y_\alpha := \sum_{i \in B_\alpha} X_i. \tag{F.1}$$

Then $A_N = \sum_\alpha Y_\alpha$. By translation covariance (A1), the $Y_\alpha$ form an approximately translation-invariant family on the coarse-grained lattice. The exponential decay of cumulants (A3) is preserved under block summation. Indeed, for any ordered tuple of blocks $(\alpha_1, \ldots, \alpha_m)$,

$$\kappa_m(Y_{\alpha_1}, \ldots, Y_{\alpha_m}) = \sum_{i_1 \in B_{\alpha_1}} \cdots \sum_{i_m \in B_{\alpha_m}} \kappa_m(X_{i_1}, \ldots, X_{i_m}).$$

Using (A3) and summing over the block volumes, we obtain

$$|\kappa_m(Y_{\alpha_1}, \ldots, Y_{\alpha_m})| \leq C_m \, |\ell|^m \, e^{-\mu \, \mathrm{dist}(\alpha_1, \ldots, \alpha_m)}, \tag{F.2}$$

where dist is the distance between blocks. Hence, after rescaling the cumulant constants, the coarse-grained variables $Y_\alpha$ satisfy the same Lieb-Robinson type exponential clustering as the original $X_i$. Similarly, the quasi-locality bound (A4) remains valid for the block variables, with a modified decay rate.

Formally, we fix $\ell$ and define the coarse-grained system on the quotient lattice $\Lambda/\ell$ with operators $Y_\alpha$. By the stability of the cumulant bounds, the normalized sum $A_N/\sqrt{N}$ has the same Gaussian limit as the original model. Thus the mapping

$$\mathcal{R}_\ell : \quad \{X_i\}_{i \in \Lambda} \longmapsto \{Y_\alpha\}_{\alpha \in \Lambda/\ell}$$

is a renormalization group (RG) transformation that preserves the Gaussian fixed point (GFP). This is the analogue of the Ising decimation map. However, whereas in the Ising model the transformation acts on scalar coupling constants, here it acts on the space of local operator families satisfying (A1)–(A3). The fixed point is characterized by the Gaussian moment relations

$$\kappa_m = 0 \quad \text{for } m \geq 3, \qquad \kappa_4 = 0, \qquad \mu_4 = 3\mu_2^2,$$

where $\mu_m$ denotes the $m$-th moment of the limiting distribution.

If a conserved charge $Q$ is present and sector assumptions (C1)–(C4) hold, the same construction applies within each sector $\mathcal{H}_q$. The coarse-grained variables $Y_\alpha^q := \sum_{i \in B_\alpha} X_i$ remain well-defined on $\mathcal{H}_q$, and the sector cumulant decay (C2) is stable under block summation in exactly the same way, yielding an analogue of (F.2) with $\tau_q$. Consequently, the decimation mapping preserves the Gaussian limit in every sector, and after averaging over sectors the global limit

remains Gaussian.

The main difference from the classical Ising case is that $\mathcal{R}_\ell$ is not exact at finite $\ell$. Since the $X_i$ do not commute, the block variables $Y_\alpha$ generate effective multi-operator corrections via nested commutators. These corrections are controlled by (A4) and are exponentially small in the block separation, hence the decimation map is an asymptotic RG transformation, valid in the limit of large separation, rather than an exact algebraic identity. In models with all-to-all interactions, such as the Johnson model, the cumulant decay fails, the block variables acquire long-range correlations, and the Gaussian fixed point is no longer stable.

## F.2 Row-sum covariance under decimation

We now examine the distribution of connectivity under decimation. With the same block partition, define the centred block row-sum function

$$U_\alpha(x) := \sum_{i \in B_\alpha} T_i(x), \tag{F.3}$$

where $T_i$ is the centred local row-sum from (A5). Then the total centred row sum is $\widetilde{R}_N(x) = \sum_\alpha U_\alpha(x)$. The classical cumulants of the block variables $U_\alpha$ satisfy

$$\kappa_m^{\mathrm{cl}}(U_{\alpha_1}, \dots, U_{\alpha_m}) = \sum_{i_1 \in B_{\alpha_1}} \cdots \sum_{i_m \in B_{\alpha_m}} \kappa_m^{\mathrm{cl}}(T_{i_1}, \dots, T_{i_m}),$$

and using (A5) we obtain

$$\left| \kappa_m^{\mathrm{cl}}(U_{\alpha_1}, \dots, U_{\alpha_m}) \right| \le C_m \, |\ell|^m \, e^{-\mu \, \mathrm{dist}(\alpha_1, \dots, \alpha_m)}. \tag{F.4}$$

Thus the block functions $U_\alpha$ satisfy the same exponential clustering as the original $T_i$, with rescaled constants. By Proposition 3.1, the normalized block row sum

$$\frac{1}{\sqrt{|\Lambda/\ell|}} \sum_\alpha U_\alpha(x)$$

converges weakly to a centred Gaussian. Since $|\Lambda/\ell| = N/\ell^2$, we have

$$\frac{1}{\sqrt{|\Lambda/\ell|}} \sum_\alpha U_\alpha(x) = \ell \, \frac{\widetilde{R}_N(x)}{\sqrt{N}},$$

so the decimated connectivity has a distribution equivalent to that of the original one up to the deterministic factor $\ell$. The limiting variance scales as $\ell^2 \, \sigma_{\mathrm{row}}^2$. In the presence of a conserved charge, the sector block functions $U_\alpha^q$ satisfy an analogous bound to (F.4) with (C4), and the same conclusion holds within each sector.

# G Approximate conservation and spectral shift

We now consider the situation where the conservation law is only approximate: $[Q, A_N] \neq 0$. We show that under a mild and physically natural condition, the Gaussian limit is preserved up to a possible shift.

## G.1 Locality of the conserved quantity

Throughout this appendix we assume that $Q$ is a sum of local terms,

$$Q = \sum_{x \in \Lambda} q_x,$$

where each $q_x$ is supported on a finite region around $x$ and has integer spectrum. Typical examples are total particle number $N = \sum_x n_x$ or total magnetization $S^z = \sum_x S^z_x$. In particular, the spectral gap of $Q$ satisfies

$$\Delta_Q \geq 1,$$

uniformly in the system size $N$. This fact is essential because it prevents the energy denominators in perturbation theory from becoming arbitrarily small as $N \to \infty$.

Define the block-diagonal part of $A_N$ with respect to $Q$:

$$A_{\mathrm{D}} := \sum_q P_q A_N P_q,$$

where $P_q$ is the projection onto the eigenspace of $Q$ with eigenvalue $q$. The off-diagonal part is

$$A_{\mathrm{OD}} := A_N - A_{\mathrm{D}}.$$

By construction $[A_{\mathrm{D}}, Q] = 0$, so $A_{\mathrm{D}}$ falls within the setting of the main theorem, provided the sector cumulant assumptions (C1)–(C3) hold for $A_{\mathrm{D}}$.

## G.2 Gaussian limit with a possible shift

We assume that the commutator is small in the following local sense:

$$\|[Q, A_N]\| = o(\sqrt{N}),$$

and that $[Q, A_N]$ is a sum of local terms with exponentially decaying tails. The second condition is automatic if both $Q$ and $A_N$ are built from local operators with finite range, by the Lieb–Robinson bounds already used in the main text.

**Remark G.1.** Since $Q$ has spectral gap $\Delta_Q \geq 1$ and $A_{\mathrm{OD}}$ is the off-diagonal part with respect to $Q$, standard perturbation theory gives

$$\|A_{\mathrm{OD}}\| \leq C \, \frac{\|[Q, A_N]\|}{\Delta_Q} \leq C \, \|[Q, A_N]\| = o(\sqrt{N}).$$

Hence

$$\frac{\|A_{\mathrm{OD}}\|}{\sqrt{N}} \to 0.$$

In particular, for every fixed integer $m \geq 1$,

$$|\tau(A_{\mathrm{OD}}^m)| \leq \|A_{\mathrm{OD}}\|^m = o(N^{m/2}).$$

Since $\|A_{\mathrm{OD}}/\sqrt{N}\| \to 0$, the spectral distribution of

$$\frac{A_N}{\sqrt{N}} = \frac{A_{\mathrm{D}}}{\sqrt{N}} + \frac{A_{\mathrm{OD}}}{\sqrt{N}}$$

is asymptotically the same as that of $A_{\mathrm{D}}/\sqrt{N}$. Under assumptions (C1)–(C3) for $A_{\mathrm{D}}$ in each sector, the main theorem implies that $A_{\mathrm{D}}/\sqrt{N}$ converges weakly to a Gaussian law with variance $\sigma^2$. Therefore $A_N/\sqrt{N}$ also converges to a Gaussian law, possibly with a shifted mean given by

$$\mu = \lim_{N\to\infty} \frac{\tau(A_{\mathrm{OD}})}{\sqrt{N}},$$

provided this limit exists. If $\tau(A_{\mathrm{OD}}) = o(\sqrt{N})$, the shift is zero.

The present setting is physically relevant when the violation of conservation is weak and local, e.g. due to a small non–number-conserving perturbation of finite range. In models with all-to-all couplings, such as the Johnson model, the commutator $\|[Q, A_N]\|$ is typically of order $O(N)$ rather than $o(\sqrt{N})$, and the sector cumulants do not decay with distance; consequently the Gaussian limit fails. This shows that locality of both the operators and the conservation-breaking terms is essential.

# References


[1] Dimitrios Ampelogiannis and Benjamin Doyon. Clustering of higher order connected correlations in $C^*$ dynamical systems. *Journal of Mathematical Physics*, 66(5):053507, 2025.

[2] Greg W. Anderson, Alice Guionnet, and Ofer Zeitouni. *An Introduction to Random Matrices*, volume 118 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge, 2010.

[3] Erwin Bolthausen. On the central limit theorem for stationary mixing random fields. *Annals of Probability*, 10(4):1047–1050, 1982.

[4] Xin Cao, Feng Wang, and Yilong Han. Ground-state phase-space structures of two-dimensional $\pm j$ spin glasses: A network approach. *Physical Review E*, 91(6):062135, 2015.

[5] Nigel Goldenfeld. *Lectures on Phase Transitions and the Renormalization Group*, volume 85 of *Frontiers in Physics*. Addison-Wesley, Reading, MA, 1992.

[6] Yilong Han. Phase-space networks of geometrically frustrated systems. *Physical Review E*, 80(5):051102, 2009.

[7] Yilong Han. Phase-space networks of the six-vertex model under different boundary conditions. *Physical Review E*, 81(4):041118, 2010.

[8] Wassily Hoeffding. A class of statistics with asymptotically normal distribution. *Annals of Mathematical Statistics*, 19(3):293–325, 1948.

[9] Elliott H. Lieb and Derek W. Robinson. The finite group velocity of quantum spin systems. *Communications in Mathematical Physics*, 28(3):251–257, 1972.

[10] Alexandru Nica and Roland Speicher. *Lectures on the Combinatorics of Free Probability*, volume 335 of *London Mathematical Society Lecture Note Series*. Cambridge University Press, Cambridge, 2006.

[11] Yi Peng, Feng Wang, K. Y. Michael Wong, and Yilong Han. Self-similarity of phase-space networks of frustrated spin models and lattice gas models. *Physical Review E*, 84(5):051105, 2011.

[12] Linh V. Tran, Van H. Vu, and Ke Wang. Sparse random graphs: Eigenvalues and eigenvectors. *Random Structures & Algorithms*, 42(1):110–134, 2013.

[13] Dan V. Voiculescu, Kenneth J. Dykema, and Alexandru Nica. *Free Random Variables*, volume 1 of *CRM Monograph Series*. American Mathematical Society, Providence, RI, 1992.

[14] Eugene P. Wigner. Characteristic vectors of bordered matrices with infinite dimensions. *Annals of Mathematics*, 62(3):548–564, 1955.

[15] Kenneth G. Wilson and John Kogut. The renormalization group and the $\epsilon$ expansion. *Physics Reports*, 12(2):75–199, 1974.